

Scalable Parallel-in-Time Integration

Acoustics in Expanding Volumes

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Work in Progress

- This is a work in progress
- Should be finished and available in the Spring!

The Equation of Motion and Causality

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- Methods of solving the EoM have been sequential
- Calculations haven't been able to benefit from parallelism
- What if they could?

Acoustics in Expanding Volumes

- Wave $\psi(t, \mathbf{x})$ defined by its density n and phase θ with

$$\psi(t, \mathbf{x}) = \sqrt{n_0 + \Delta n(t, \mathbf{x})} e^{i(\theta_0 + \Delta\theta(t, \mathbf{x}))}$$

- Wave equation for an expanding volume:

$$\partial_t^2 \theta - \frac{\dot{a}}{a} \partial_t \theta - ac_0^2 \nabla^2 \theta = 0$$

- Spectral decomposition in space: $\tilde{\theta}_{\mathbf{k}} = \mathcal{F}(\Delta\theta)(t, \mathbf{k})$ for $k < k_c(t)$

The Field Equation

$$\partial_t^2 \tilde{\theta}_{\mathbf{k}} - \frac{\dot{a}}{a} \partial_t \tilde{\theta}_{\mathbf{k}} - ac_0^2 k^2 \tilde{\theta}_{\mathbf{k}} = 0$$

$$\tilde{\theta}_{\mathbf{k}}(0) = \tilde{\theta}_{\mathbf{k}0} \quad \partial_t \tilde{\theta}_{\mathbf{k}}(0) = -U_0 n_{\mathbf{k}0} / h$$

Acoustics in Expanding Volumes

$$\partial_t^2 \theta - \frac{\dot{a}}{a} \partial_t \theta - ac_0^2 \nabla^2 \theta = 0 \quad \xRightarrow{\tilde{\theta}_{\mathbf{k}} = \mathcal{F}(\Delta \theta)(t, \mathbf{k})}$$

The Parareal Algorithm

The 3 cores steps of the Parareal Algorithm

1. Prepare the subproblems
2. Propagate each subproblem in parallel
3. Correct

The Parareal Algorithm - Subproblem Preparation

1. Choose number of subproblems
(Suggest number of cores)
- 2.

The Parareal Algorithm - Parallel Propagation

The Parareal Algorithm - Corrections

GPU Computing

- Each subproblem can be computed in parallel on each gpu core

Distributed Computing

- 1 wave number per gpu
- many machines with gpus can further parallelize the calculation

Some Basic Results

- pictures of results

Numerical Analysis

- Error
- Convergence
- Stability

Algorithm Analysis

- Time Complexity
- Space Complexity

Conclusion

- Accessibility to parallelization is increasing
- Problems only becoming more complex

Thank you for your time.

- Code available on GitHub →
- Let's collaborate!
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